Passive Network — AI-Powered Passive Network Detection & Revenue Recovery

Ontology × Network Topology × AI = Proactive Detection

Powered by NexusAI

Sub-Saharan Africa Telco
~1,000 sites · 2.3M subscribers

PROJECTED ROI

Proactive Detection, Revenue Protection,
MTTR Reduction and Dispatch Optimization

40%
MTTR reduction

$3.15M–$4.75M
Total annual impact

HOW PASSIVE NETWORK WORKS

DATA SOURCES

- **RAN Performance Files** — Ericsson 2G cell metrics
- **OSS Alarm Streams** — Real-time fault events
- **ENIQ Data Warehouse** — 85,000+ tables, 14-day rolling window
- **BSS Subscriber Records** — Revenue & usage data

Passive Network AI Engine

Ontology-driven reasoning over Cell→Neighbor→Transmission→BSC chains

- Ontology Mapping
- Anomaly Detection
- Neighbor Analysis
- Root-Cause ML
- Revenue Metering
- Auto-Remediation

INSIGHTS DELIVERED

- **Detection Alerts** — Confidence score & reason
- **Revenue Impact** — $ per hour per incident
- **Root-Cause Reports** — Failure category & evidence
- **Action Plans** — Tiered remediation steps

70%+	40%	4-6 hrs	30-40%	90 days
Passive network cells detected proactively	MTTR reduction achieved	Detection time eliminated	Unnecessary dispatches avoided	Full deployment, no lock-in

THE CHALLENGE

- **Reactive Detection** — Passive network cells found only after customer complaints pile up — 4-6 hours after degradation begins.
- **Data Fragmentation** — Intelligence scattered across Ericsson RAN, OSS, ENIQ, AUTIN and BSS — no unified view exists.
- **No Revenue Awareness** — Faults have no link to financial impact — NOC works by queue order, not business value.
- **Tribal Knowledge Detection** — Engineers manually query for 100% availability + zero traffic and review trends by eye.
- **Avoidable Field Dispatches** — 30-40% of truck rolls are unnecessary — root cause could be verified remotely first.
- **Slow Resolution Cycles** — 8-24 hour MTTR depending on complexity, with no learning loop to accelerate future fixes.

THE SOLUTION

An ontology-first AI layer deployed in 90 days on the operator's live Ericsson 2G network — detecting passive network cells within minutes, calculating real-time revenue impact, and auto-remediating where safe. No systems replaced.

Ontology-Driven Detection
Continuously monitors all 1,000 cells via Cell→Neighbor→Transmission→BSC relationship chains. Flags passive conditions within minutes.

Neighbor Overflow Analysis
Queries adjacent cells for traffic spikes confirming load redistribution — correlates active alarms on transmission links.

Root-Cause Identification
Traverses ontology to identify failure category (transmission link failure vs. software fault) with quantified confidence level.

Real-Time Revenue Metering
Calculates affected subscribers, revenue loss per hour, and projected total loss if unresolved for historical average MTTR.

Automated Remediation
Executes soft-resets where ontology confirms safety. Escalates to L2 or field dispatch only when physical fault evidence requires it.

NOC Dashboards & Knowledge Transfer
Real-time cell health map, active alert queue, ontology graph visualization, and operational metrics with AI reasoning chain.

RESULTS DELIVERED

70%+ Proactive Detection
Passive network cells detected before any customer complaint — vs. 0% pro
detection before.

40% MTTR Reduction
AI-guided investigation workflows replace hours of manual data-gathering a
siloed systems.

50% Investigation Time Cut
Automated root-cause analysis in seconds via ontology traversal vs. manual
system review.

$2.6M–$3.9M Revenue Protected
Annual revenue at risk from passive network cells now monitored and defen
real time.

30–40% Dispatch Reduction
Remote verification confirms root cause before sending technicians —
$150K-$250K saved annually.

90 Days Full Production
Entire capability live in production within the engagement window — no ve
lock-in.

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